Probabilistic matching: Fellegi–Sunter on what the rules left

For the records the deterministic stage left unresolved:

1. **Candidates:** generate pairs against all records (exact keys, sorted neighbourhood, TF-IDF).
2. **Scoring:** Splink's **Fellegi–Sunter** model scores exactly those pairs. It is fitted on train: λ from near-certain rules, u by random sampling, m by EM. A score doesn't depend on the size of the scored set: term frequencies come from train, and λ is rescaled to the scored set's size.
3. **Decision:** accept at `match_weight ≥` a threshold.
4. **Resolution:** cluster with the deterministic pairs.

How configurations are chosen, without overfitting:

- **The threshold** comes from one rule on **out-of-fold** train scores (entity k-fold CV), for every configuration: `SPEC`, each decision variant, each Optuna trial.
- **Variants** are adopted only by a rule (`adopt`): a paired gain on cross-validated train and no loss on val. Optuna's suggestion is judged by the same rule.
- **test** is scored once.

The configuration flows `SPEC` (as written) → `TUNED` (threshold set) → `FINAL` (decisions applied); no cell reassigns them.

Cells: Configuration · Setup · Design analysis (train) · Training · Candidates (val) · Threshold · A/B (val, paired) · Decisions · Evaluation (val) · Output (train / val / test → artifacts, gold) · Optuna + Pareto.

In [1]:
```
import json
import time
from dataclasses import replace

import optuna
import pandas as pd
from IPython.display import display

from er.candidates import estimate
```

```
from er.config import (Budget, Comparator, Comparison, FSModel, Key, KeyBlocker, Level, OverBudget, ProbSpec, Prob
Tune, Resolver,
                       SortedNeighbourhood, TfidfNeighbours, ThresholdRule, with_level, without_tf)
from er.evaluate import adopt, c2, candidate_metrics, missed_pairs, paired_test, predicted_pairs, true_pairs
from er.io import append_leaderboard, load, read_stage, test_evaluations, write_gold, write_stage
from er.match import parameters, train_fs, u_values, waterfall
from er.pipeline import (ProbResult, accept_curve, cross_validate_fs, cv_threshold_curve, entity_folds, run_fs, se
lect_threshold, summed,
                         threshold_curve)
from er.settings import load_settings
from er.standardize import prepare
from er.strings import connect
from er.tune import new_study, pick_within_noise, plain, start_dashboard
```

Configuration

The only cell, with `config/settings.toml`, that names columns. Correlated columns share one comparison (address code / normalized address / postal form *address*), because Fellegi–Sunter assumes independence.

In [2]:
```
S = load_settings()
DATA = S.data
FOLD, COMPANY = ("kana_fold",), ("nfkc", "strip_legal_form", "kana_fold")

GENERATORS = (
    KeyBlocker(name="phone", keys=(Key("PHONE_NUM_4"),)),
    KeyBlocker(name="address", keys=(Key("ADDRESS", ("address",)),)),
    KeyBlocker(name="address_code", keys=(Key("SOURCE_ADDR_CD"),)),
    KeyBlocker(name="cif", keys=(Key("CIF_ID"),)),
    KeyBlocker(name="person_name", keys=(Key("PERSON_NAME", FOLD),)),
    KeyBlocker(name="company_name", keys=(Key("NYS_CNV_P_NAME_KANA", COMPANY),)),
    SortedNeighbourhood(name="sn_person_name", key=Key("PERSON_NAME", FOLD), window=4),
    SortedNeighbourhood(name="sn_company_name", key=Key("NYS_CNV_P_NAME_KANA", COMPANY), window=4),
    TfidfNeighbours(name="tfidf_person_name", key=Key("PERSON_NAME", FOLD), ngram=(2, 3), k=5, min_cosine=0.5),
    TfidfNeighbours(name="tfidf_company_name", key=Key("NYS_CNV_P_NAME_KANA", COMPANY), ngram=(2, 3), k=5, min_cos
ine=0.5),
)

SPEC = ProbSpec(
```

```
    standardize=S.standardize,
    previous_stage="deterministic",
    stage="probabilistic",
    blockers=GENERATORS,
    model=FSModel(
        name="fs",
        comparisons=(
            Comparison(name="name", null_if_missing=("PERSON_NAME",), tf=Key("PERSON_NAME", FOLD), levels=(
                Level(label="exact", any_of=(("PERSON_NAME", Comparator("exact")),), tf=True),
                Level(label="fold exact", any_of=(("PERSON_NAME", Comparator("exact", normalize=FOLD)),), tf=Tru
e),
                Level(label="fold JW high", any_of=(("PERSON_NAME", Comparator("jaro_winkler", 0.92, FOLD)),)),
                Level(label="given name", any_of=(("GIVEN_NAME_KANA", Comparator("exact", normalize=FOLD)),)),  #
surname change
                Level(label="fold JW low", any_of=(("PERSON_NAME", Comparator("jaro_winkler", 0.85, FOLD)),)),
            )),
            Comparison(name="company", null_if_missing=("NYS_CNV_P_NAME_KANA",), tf=Key("NYS_CNV_P_NAME_KANA", COM
PANY), levels=(
                Level(label="exact", any_of=(("NYS_CNV_P_NAME_KANA", Comparator("exact", normalize=COMPANY)),), tf
=True),
                Level(label="JW high", any_of=(("NYS_CNV_P_NAME_KANA", Comparator("jaro_winkler", 0.92, COMPAN
Y)),)),
                Level(label="JW low", any_of=(("NYS_CNV_P_NAME_KANA", Comparator("jaro_winkler", 0.85, COMPAN
Y)),)),
            )),
            Comparison(name="address", null_if_missing=("SOURCE_ADDR_CD", "ADDRESS", "POSTAL"), levels=(
                Level(label="same address", any_of=(("SOURCE_ADDR_CD", Comparator("exact")),
                                                    ("ADDRESS", Comparator("exact", normalize=("address",))))),
                Level(label="address close", any_of=(("ADDRESS", Comparator("levenshtein", 2, ("address",))),)),
                Level(label="same postal", any_of=(("POSTAL", Comparator("exact")),)),
            )),
            Comparison(name="phone", null_if_missing=("PHONE_NUM_4",), levels=(
                Level(label="exact", any_of=(("PHONE_NUM_4", Comparator("exact")),)),
                Level(label="one digit off", any_of=(("PHONE_NUM_4", Comparator("hamming", 1)),)),
            )),
            Comparison(name="cif", null_if_missing=("CIF_ID",), levels=(
                Level(label="exact", any_of=(("CIF_ID", Comparator("exact")),)),
            )),
        ),
        prior_rules=((Key("CIF_ID"), Key("FULL_NAME_KANA")), (Key("CIF_ID"), Key("NYS_CNV_P_NAME_KANA")),
                     (Key("FULL_NAME_KANA"), Key("SOURCE_ADDR_CD"), Key("PHONE_NUM_4"))),
```

```
        prior_recall=0.5,  # those rules recall about half of the true pairs (deterministic exact stage, measured)
        u_max_pairs=1e7,
        m_method="em",
        em_rules=((Key("PHONE_NUM_4"),), (Key("ADDRESS", ("address",)),)),
        seed=0,
    ),
    min_match_weight=4.0,  # starting value; THRESHOLD_RULE sets the real one (Threshold cell)
    resolver=Resolver(method="connected_components"),
    budget=Budget(max_candidate_pairs=5_000_000, max_block_size=5_000, target_rows=1_000_000, benchmark_pairs=20_0
00),
)

CV_FOLDS, CV_SEED = 3, 0
THRESHOLD_RULE = ThresholdRule(grid=(0, 2, 4, 6, 8, 10, 12, 14, 16), max_fp=0, max_fp_rate_hi=0.05, margin_steps=
1)



def optuna_suggestion(s: ProbSpec) -> ProbSpec:  # an Optuna study's suggestion (trial 12); passed adopt vs FINAL
    model = without_tf(s.model)
    for (comparison, label), threshold in {("name", "fold JW high"): 0.95, ("name", "fold JW low"): 0.75,
                                           ("company", "JW high"): 0.97, ("company", "JW low"): 0.80}.items():
        model = with_level(model, comparison, label, threshold=threshold)
    off = {"person_name", "company_name", "sn_person_name", "tfidf_company_name"}
    return replace(s, blockers=tuple(b for b in GENERATORS if b.name not in off), model=model)


CHANGES = {  # each variant, defined once as a change to a configuration
    "labels": lambda s: replace(s, model=replace(s.model, m_method="labels")),
    "no_term_frequency": lambda s: replace(s, model=without_tf(s.model)),
    "exact_keys_only": lambda s: replace(s, blockers=tuple(b for b in s.blockers if isinstance(b, KeyBlocker))),
    "levenshtein_names": lambda s: replace(s, model=with_level(with_level(s.model, "name", "fold JW high", functio
n="levenshtein",
                                                                          threshold=1), "name", "fold JW low", fun
ction="levenshtein", threshold=2)),
    "constrained_given_name": lambda s: replace(s, resolver=Resolver(method="constrained", cannot_link=("GIVEN_NAM
E_KANA",))),
    "optuna_suggestion": optuna_suggestion,
    "u_from_1e6_pairs": lambda s: replace(s, model=replace(s.model, u_max_pairs=1e6)),  # u-sampling ≈ 6× cheaper
}
# adopted in order, each on top of the choice so far: "better" needs a significant paired gain on cross-validated
```

```
train
# (p < DECISION_P, no FP added); "cheaper" needs no paired loss there. Both need no paired loss on val (er.evaluat
e.adopt).
DECISIONS = (("no_term_frequency", "better"), ("exact_keys_only", "cheaper"), ("optuna_suggestion", "better"),
             ("u_from_1e6_pairs", "cheaper"))
DECISION_P = 0.05

# the worked example ("Why a pair matches"): columns shown; the example match has these filled on both sides, the
first differing
EXPLAIN_SHOW = ("FULL_NAME_KANA", "NYS_CNV_P_NAME_KANA", "ADDRESS", "PHONE_NUM_4", "CIF_ID")
EXPLAIN_FILLED = ("FULL_NAME_KANA", "ADDRESS", "CIF_ID")

PTUNE = ProbTune(
    toggle_blockers=("person_name", "company_name", "sn_person_name", "sn_company_name", "tfidf_person_name", "tfi
df_company_name"),
    m_methods=("em", "labels"),
    tf_options=(True, False),
    level_thresholds={("name", "fold JW high"): (0.88, 0.97, 0.01), ("name", "fold JW low"): (0.75, 0.90, 0.01),
                      ("company", "JW high"): (0.88, 0.97, 0.01), ("company", "JW low"): (0.75, 0.90, 0.01)},
    resolvers={"connected_components": Resolver(method="connected_components"),
               "constrained_given_name": Resolver(method="constrained", cannot_link=("GIVEN_NAME_KANA",))},
    n_trials=30, seed=0,
    storage_path=S.paths["optuna"] / "probabilistic_matching.db", study_prefix="probabilistic",
    dashboard_host="127.0.0.1", dashboard_port=8081,  # 8080 belongs to the deterministic notebook
)
```

Setup

Three caches skip repeated work without changing any result:

- **Models,** by (config, fitting set).
- **u, per comparison,** by `FSModel.u_keys()`: a comparison's u depends only on itself and the fitting records, so every model that shares a comparison reuses its u (identical results; see tests/). Optuna trials mostly change name and company thresholds, so address and phone u are sampled once per fold.
- **Runs,** by (config, dataset, threshold).

In [3]:

```
CON, DB = connect()  # DuckDB with character-level cl_* string functions
uid = DATA.unique_id
l, r = f"{uid}_l", f"{uid}_r"
SHOW = ["stage", "TP", "FP", "recall", "recall_lo", "recall_hi", "fp_rate_hi", "pairs_FP", "overmerged", "dup_unre
solved"]
train_records, train_truth = load(DATA, "train")
train_prev_pairs, train_prev_resolved = read_stage(DATA, "train", SPEC.previous_stage)
FOLDS = entity_folds(train_truth, DATA, CV_FOLDS, CV_SEED)
MODELS, U, RUNS, CVS = {}, {}, {}, {}


def fit(spec: ProbSpec, records: pd.DataFrame, truth: pd.DataFrame, key: object) -> dict:
    if (repr(spec.model), key) not in MODELS:
        df = prepare(spec.standardize, records, spec.normalized(), spec.used())
        u_keys = spec.model.u_keys()
        model_json = train_fs(spec.model, df, DB, uid, labels=truth.set_index(uid)[DATA.entity_id],
                              u={c: U[k, key] for c, k in u_keys.items() if (k, key) in U})
        for c, u in u_values(model_json).items():
            U.setdefault((u_keys[c], key), u)
        MODELS[repr(spec.model), key] = model_json
    return MODELS[repr(spec.model), key]


def run(spec: ProbSpec, dataset: str) -> ProbResult:  # model fitted on all of train
    key = (repr(spec), dataset)
    if key not in RUNS:
        records, truth = load(DATA, dataset)
        RUNS[key] = run_fs(spec, fit(spec, train_records, train_truth, "train"), records, truth,
                           *read_stage(DATA, dataset, spec.previous_stage), DB, DATA)
    return RUNS[key]


def cv(spec: ProbSpec) -> list[ProbResult]:  # out-of-fold results on train
    if repr(spec) not in CVS:
        CVS[repr(spec)] = cross_validate_fs(spec, fit, train_records, train_truth, train_prev_pairs, train_prev_re
solved, FOLDS, DB, DATA)
    return CVS[repr(spec)]


def threshold_of(spec: ProbSpec) -> float:  # THRESHOLD_RULE on this configuration's out-of-fold scores (no resolu
```

```
tion needed)
    return select_threshold(accept_curve(cv(spec), list(THRESHOLD_RULE.grid)), THRESHOLD_RULE)[0]


def joined(results: ProbResult | list[ProbResult]) -> set[tuple[str, str]]:  # every pair the clusters put togethe
r
    return set().union(*(predicted_pairs(x.records.set_index(uid)["CLUSTER_ID"]) for x in (results if isinstance(r
esults, list) else [results])))


def truth_set(truth: pd.DataFrame) -> set[tuple[str, str]]:
    return set(map(tuple, true_pairs(truth.set_index(uid)[DATA.entity_id], uid).to_numpy()))
```

Design analysis on train

Which keys reach the true pairs the deterministic stage missed, and how many different-entity pairs each key would also produce. This justifies the candidate keys; it uses train only.

In [4]:
```
t_df = prepare(SPEC.standardize, train_records, SPEC.normalized(), SPEC.used())
t_entity = train_truth.set_index(uid)[DATA.entity_id]
t_prev = train_prev_resolved.set_index(uid)["cluster_id"].reindex(t_df[uid].to_numpy())
targets = missed_pairs(t_prev, t_entity.reindex(t_prev.index), uid)
side, keyed_rows = t_df.set_index(uid), t_df.assign(__entity=t_df[uid].map(t_entity))
rows = []
for b in (g for g in GENERATORS if isinstance(g, KeyBlocker)):
    cols = [k.name for k in b.keys]
    agree = (side.loc[targets[l], cols].reset_index(drop=True) == side.loc[targets[r], cols].reset_index(drop=Tru
e)).fillna(False).all(axis=1)
    keyed = keyed_rows.dropna(subset=cols)
    rows.append(dict(key=b.name, missed_pairs_agree=float(agree.mean()),
                     different_entity_pairs=c2(keyed.groupby(cols).size()) - c2(keyed.groupby(cols + ["__entit
y"]).size())))
print(f"train: {len(targets):,} true pairs missed by the deterministic stage")
display(pd.DataFrame(rows).round(4))
```

```
train: 2,135 true pairs missed by the deterministic stage
```

	key	missed_pairs_agree	different_entity_pairs
0	phone	0.5073	45506
1	address	0.3789	11504
2	address_code	0.3438	10671
3	cif	0.1077	0
4	person_name	0.3878	371
5	company_name	0.0806	386

Training on train: EM vs labels

The weight of a level is log2(m/u): how much more often true matches land there than random pairs. EM is unsupervised (usable on real data); labels is the supervised reference. A level without an estimate never uses Splink's defaults: no sampled pair → u at its 95% upper bound; never seen in training → m = u (0 bits). Those levels are listed.

In [5]:
```
n, k = len(t_df), len(SPEC.model.comparisons)
em_p = sum(estimate(KeyBlocker(name="em", keys=rule), t_df, n)["est_pairs"] for rule in SPEC.model.em_rules)
print(f"cost: u O(max_pairs·C) = O({SPEC.model.u_max_pairs:,.0f}·{k}) · EM O(iterations·P·C), P = {em_p:,} · label
s O(Σ k² per entity)")
DATA.models_dir.mkdir(parents=True, exist_ok=True)
weights = {}
for method in ("em", "labels"):
    spec, t = replace(SPEC, model=replace(SPEC.model, m_method=method)), time.perf_counter()
    model_json = fit(spec, train_records, train_truth, "train")
    (DATA.models_dir / f"{spec.model.name}_{method}.json").write_text(json.dumps(model_json, ensure_ascii=False, i
ndent=1))
    weights[method] = parameters(model_json).set_index(["comparison", "level"])[["m", "u", "weight"]]
    filled = model_json["er"]["filled"]
    print(f"{method}: fitted in {time.perf_counter() - t:.1f}s · no estimate: u {filled['u'] or '–'} · m {filled
['m'] or '–'}")
display(pd.concat(weights, axis=1).round(3))
```

```
cost: u O(max_pairs·C) = O(10,000,000·5) · EM O(iterations·P·C), P = 66,351 · labels O(Σ k² per entity)
em: fitted in 8.5s · no estimate: u – · m –
labels: fitted in 0.2s · no estimate: u – · m –
```

		em			labels		
		m	u	weight	m	u	weight
comparison	level						
name	exact	0.543	0.000	15.569	0.542	0.000	15.567
	fold exact	0.218	0.000	15.532	0.218	0.000	15.530
	fold JW high	0.090	0.000	11.374	0.090	0.000	11.378
	given name	0.110	0.000	7.811	0.108	0.000	7.790
	fold JW low	0.004	0.000	3.578	0.004	0.000	3.779
	else	0.035	0.999	-4.816	0.037	0.999	-4.741
company	exact	0.654	0.000	13.126	0.593	0.000	12.984
	JW high	0.012	0.000	9.511	0.010	0.000	9.194
	JW low	0.003	0.000	4.452	0.003	0.000	4.439
	else	0.331	1.000	-1.596	0.395	1.000	-1.341
address	same address	0.708	0.000	14.015	0.708	0.000	14.014
	address close	0.004	0.000	8.893	0.003	0.000	8.595
	same postal	0.113	0.000	10.823	0.114	0.000	10.839
	else	0.175	1.000	-2.515	0.175	1.000	-2.515
phone	exact	0.746	0.000	12.540	0.745	0.000	12.539
	one digit off	0.081	0.004	4.497	0.081	0.004	4.496
	else	0.173	0.996	-2.525	0.174	0.996	-2.518
cif	exact	0.464	0.000	15.920	0.472	0.000	15.942
	else	0.536	1.000	-0.901	0.528	1.000	-0.920

Candidates on val

Each generator alone and cumulatively. `completeness_on_targets` is the share of the deterministic stage's missed true pairs that the candidates contain, i.e. the ceiling for this stage.

In [6]:
```
val_records, val_truth = load(DATA, "val")
res = run(SPEC, "val")
display(res.report[["generator", "big_o", "filled", "pairs", "completeness_on_targets", "seconds"]].round(4))
prev = read_stage(DATA, "val", SPEC.previous_stage)[1].set_index(uid)["cluster_id"]
entity = val_truth.set_index(uid)[DATA.entity_id]
val_targets = missed_pairs(prev, entity.reindex(prev.index), uid)
found_by = res.candidates["sources"].str.split("+")
display(pd.DataFrame([dict(up_to=b.name, **{key: v for key, v in candidate_metrics(
    res.candidates[found_by.map(lambda s, names={g.name for g in GENERATORS[: i + 1]}: bool(names & set(s)))], ent
```

```
ity, uid, val_targets).items()
    if key in ("candidates", "completeness_on_targets")}) for i, b in enumerate(GENERATORS)]).round(4))
```

	generator	big_o	filled	pairs	completeness_on_targets	seconds
0	phone	O(N + P)	O(10,002 + 6,634)	4985	0.4758	0.019
1	address	O(N + P)	O(10,002 + 2,540)	1380	0.3690	0.023
2	address_code	O(N + P)	O(10,002 + 2,350)	1277	0.3133	0.020
3	cif	O(N + P)	O(10,002 + 1,041)	81	0.1186	0.020
4	person_name	O(N + P)	O(10,002 + 1,388)	318	0.3953	0.019
5	company_name	O(N + P)	O(10,002 + 217)	116	0.0966	0.017
6	sn_person_name	O(N log N + N·w)	O(7,909·log 7,909 + 7,909·4)	20991	0.4261	0.039
7	sn_company_name	O(N log N + N·w)	O(1,640·log 1,640 + 1,640·4)	4683	0.0966	0.031
8	tfidf_person_name	O(N·L + U·N·q)	O(7,909·L + 7,300·7,909·14)	3242	0.4510	0.360
9	tfidf_company_name	O(N·L + U·N·q)	O(1,640·L + 7,300·1,640·18)	371	0.0981	0.046

	up_to	candidates	completeness_on_targets
0	phone	4985	0.4758
1	address	6292	0.7379
2	address_code	6528	0.7877
3	cif	6566	0.8433
4	person_name	6657	0.9063
5	company_name	6724	0.9312
6	sn_person_name	27376	0.9356
7	sn_company_name	31942	0.9356
8	tfidf_person_name	33655	0.9385
9	tfidf_company_name	33871	0.9385

Threshold: chosen on cross-validated train

`THRESHOLD_RULE` : on out-of-fold train scores, the lowest weight with no false accepted pair and a stage FP upper bound ≤ `max_fp_rate_hi` , then `margin_steps` stricter. That gives `TUNED` .

In [7]:
```
t = time.perf_counter()
weight, curve = select_threshold(cv_threshold_curve(cv(SPEC), SPEC, train_prev_pairs, train_truth, FOLDS, DB, DAT
A,
                                                    list(THRESHOLD_RULE.grid)), THRESHOLD_RULE)
assert weight == threshold_of(SPEC)  # the full curve and the pair counts the rule reads pick the same weight
TUNED = replace(SPEC, min_match_weight=weight)
```

```
print(f"{CV_FOLDS}-fold CV: O(k · (fit + candidates + score + resolve)) in {time.perf_counter() - t:.1f}s → min_match_weight = {weight}")
display(curve[["min_match_weight", "accepted", "pairs_FP", "stage_fp_rate_hi", "TP", "FP", "recall", "recall_lo", "recall_hi", "eligible"]].round(4))
```

```
3-fold CV: O(k · (fit + candidates + score + resolve)) in 32.4s → min_match_weight = 8.0
```

	min_match_weight	accepted	pairs_FP	stage_fp_rate_hi	TP	FP	recall	recall_lo	recall_hi	eligible
0	0	1654	15	0.0139	6494	18	0.9360	0.9300	0.9415	False
1	2	1607	14	0.0136	6448	17	0.9294	0.9231	0.9352	False
2	4	1503	1	0.0032	6360	1	0.9167	0.9100	0.9230	False
3	6	1397	0	0.0021	6260	0	0.9023	0.8951	0.9090	True
4	8	1216	0	0.0025	6080	0	0.8763	0.8684	0.8839	True
5	10	907	0	0.0033	5765	0	0.8309	0.8219	0.8396	True
6	12	488	0	0.0061	5326	0	0.7677	0.7576	0.7774	True
7	14	451	0	0.0066	5284	0	0.7616	0.7514	0.7715	True
8	16	386	0	0.0077	5220	0	0.7524	0.7421	0.7624	True

A/B on val, paired against TUNED

Each `CHANGES` entry applied to `TUNED`, compared on the same records: true pairs gained / lost, an exact sign test (`p`), false pairs added / removed.

In [8]:
```
val_true, base, ab = truth_set(val_truth), joined(run(TUNED, "val")), []
for name, change in CHANGES.items():
    spec = change(TUNED)
    m = run(spec, "val").metrics.iloc[-1]
    row = dict(recall=m["recall"], FP=m["FP"], fp_rate_hi=m["fp_rate_hi"], **paired_test(base, joined(run(spec, "val")), val_true),
               candidates=run(spec, "val").stage_metrics["candidates"])
    append_leaderboard(DATA, f"probabilistic:{name}", spec, "val", row)
    ab.append(dict(variant=name, **row))
display(pd.DataFrame(ab).round(4))
```

	variant	recall	FP	fp_rate_hi	TP_gains	TP_losses	p	FP_added	FP_removed	candidates
0	labels	0.8864	0	0.0015	0	0	1.0000	0	0	33871
1	no_term_frequency	0.8873	0	0.0015	2	0	0.5000	0	0	33871
2	exact_keys_only	0.8864	0	0.0015	0	0	1.0000	0	0	6724
3	levenshtein_names	0.8895	0	0.0015	8	1	0.0391	0	0	33871
4	constrained_given_name	0.7705	0	0.0017	0	258	0.0000	0	0	33871
5	optuna_suggestion	0.8936	0	0.0015	19	3	0.0009	0	0	14180
6	u_from_1e6_pairs	0.8806	0	0.0015	3	16	0.0044	0	0	33871

Decisions → FINAL

Each `DECISIONS` entry is applied on top of the choice so far, at its own `THRESHOLD_RULE` threshold, and adopted only if it passes `adopt`: paired on cross-validated train, plus no paired loss on val.

In [9]:
```
train_true, FINAL, verdicts = truth_set(train_truth), TUNED, []
for name, kind in DECISIONS:
    variant = CHANGES[name](FINAL)
    variant = replace(variant, min_match_weight=threshold_of(variant))
    on_cv = paired_test(joined(cv(FINAL)), joined(cv(variant)), train_true)
    on_val = paired_test(joined(run(FINAL, "val")), joined(run(variant, "val")), val_true)
    adopted = adopt(kind, on_cv, on_val, DECISION_P)
    verdicts.append(dict(change=name, rule=kind, adopted=adopted, **{f"cv_{x}": v for x, v in on_cv.items()},
                         val_TP_gains=on_val["TP_gains"], val_TP_losses=on_val["TP_losses"], val_FP_added=on_val
["FP_added"]))
    if adopted:
        FINAL = variant
display(pd.DataFrame(verdicts).round(4))
print(f"FINAL: {len(FINAL.blockers)} generators · term frequency {'on' if any(c.tf for c in FINAL.model.comparison
s) else 'off'} · "
      f"m by {FINAL.model.m_method} · min_match_weight {FINAL.min_match_weight} · {FINAL.resolver.method}")
```

	change	rule	adopted	cv_TP_gains	cv_TP_losses	cv_p	cv_FP_added	cv_FP_removed	val_TP_gains	val_TP_losses	val_FP_added
0	no_term_frequency	better	True	15	0	0.0001	0	0	2	0	0
1	exact_keys_only	cheaper	False	0	4	0.1250	0	0	0	0	0
2	optuna_suggestion	better	True	198	0	0.0000	0	0	62	1	0
3	u_from_1e6_pairs	cheaper	False	10	24	0.0243	0	0	0	3	0

```
FINAL: 6 generators · term frequency off · m by em · min_match_weight 6.0 · connected_components
```

Evaluation on val (FINAL)

Cumulative per stage: the `deterministic` row reproduces the previous stage. `pairs_*` are matching metrics; the rest are resolution metrics. The transfer line checks the threshold, chosen on cross-validated train, against val: val's false-pair rate should stay within the CV upper bound. Then the false positives, and the true pairs just below the threshold.

In [10]:
```
res = run(FINAL, "val")
display(res.metrics[SHOW].round(4))
val_curve = threshold_curve(res, FINAL, read_stage(DATA, "val", FINAL.previous_stage)[0], val_truth, DB, DATA, lis
t(THRESHOLD_RULE.grid))
display(val_curve[["min_match_weight", "accepted", "pairs_FP", "TP", "FP", "recall", "fp_rate_hi"]].round(4))
cv_at = cv_threshold_curve(cv(FINAL), FINAL, train_prev_pairs, train_truth, FOLDS, DB, DATA, [FINAL.min_match_weig
ht]).iloc[0]
val_at = val_curve.set_index("min_match_weight").loc[FINAL.min_match_weight]
val_rate = val_at["pairs_FP"] / max(val_at["accepted"], 1)
print(f"transfer at weight {FINAL.min_match_weight}: CV {cv_at['pairs_FP']:.0f} false of {cv_at['accepted']:.0f} a
ccepted pairs "
      f"(rate ≤ {cv_at['stage_fp_rate_hi']:.2%}) · val {val_at['pairs_FP']:.0f} of {val_at['accepted']:.0f} ({val_
rate:.2%}) → "
      f"{'consistent' if val_rate <= cv_at['stage_fp_rate_hi'] else 'VAL ABOVE THE CV BOUND'}")
cols = sorted(c for c in FINAL.used() if c in res.records)
side = res.records.set_index(uid)[cols]


def side_by_side(pairs: pd.DataFrame) -> pd.DataFrame:
    return pd.concat([pairs[["match_weight", "sources"]].reset_index(drop=True), side.loc[pairs[l]].add_suffix("_
l").reset_index(drop=True),
                      side.loc[pairs[r]].add_suffix("_r").reset_index(drop=True)], axis=1).T
```

```python
accepted = res.scored["match_weight"] >= FINAL.min_match_weight
print(f"false positives accepted: {int((accepted & res.scored['fp']).sum())}")
if (accepted & res.scored["fp"]).any():
    display(side_by_side(res.scored[accepted & res.scored["fp"]].nlargest(DATA.show_rows, "match_weight")))
print("true pairs just below the threshold:")
display(side_by_side(res.scored[~accepted & ~res.scored["fp"]].nlargest(DATA.show_rows, "match_weight")))
```

	stage	TP	FP	recall	recall_lo	recall_hi	fp_rate_hi	pairs_FP	overmerged	dup_unresolved
0	deterministic	1544	0	0.6933	0.6738	0.7121	0.0019	0	0	1091
1	probabilistic	2037	0	0.9147	0.9024	0.9256	0.0015	0	0	316

	min_match_weight	accepted	pairs_FP	TP	FP	recall	fp_rate_hi
0	0	531	5	2086	5	0.9367	0.0050
1	2	519	2	2079	2	0.9335	0.0030
2	4	491	0	2058	0	0.9241	0.0015
3	6	471	0	2037	0	0.9147	0.0015
4	8	423	0	1990	0	0.8936	0.0015
5	10	300	0	1872	0	0.8406	0.0016
6	12	174	0	1730	0	0.7768	0.0017
7	14	136	0	1692	0	0.7598	0.0018
8	16	112	0	1669	0	0.7494	0.0018

```
transfer at weight 6.0: CV 0 false of 1430 accepted pairs (rate ≤ 0.21%) · val 0 of 471 (0.00%) → consistent
false positives accepted: 0
true pairs just below the threshold:
```

	0	1	2	3	4	5	6	7
match_weight	5.885354	5.885354	5.885354	5.885354	5.570724	5.567918	5.567918	5.567918
sources	phone	phone	phone	phone	sn_company_name	phone+address+address_code	phone+address+address_code	phone+address
ADDRESS_l	<NA>	岐阜県高山市八軒町3-30-18	<NA>	<NA>	大阪府大阪市生野区勝山南1-18-10	東京都調布市深大寺南町3-20-6	福岡県春日市上白水4-12-28	高槻市唐崎南１－５－７
ADDRESS__address_l	<NA>	高山市八軒町3-30-18	<NA>	<NA>	大阪市生野区勝山南1-18-10	調布市深大寺南町3-20-6	春日市上白水4-12-28	高槻市唐崎南1-5-7
CIF_ID_l	9394783785889	5838503911029	2081339529034	2096382436427	1734023849310	8407456413966	3875582213575	7927483493058
FULL_NAME_KANA_l	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>	イノウエハル	オオイシルミナ
GIVEN_NAME_KANA__kana_fold_l	<NA>	<NA>	<NA>	<NA>	<NA>	チカコ	ハル	ルミナ
NYS_CNV_P_NAME_KANA_l	カブシキガイシャゾディアックキゴウテレビセイサクカイシャキゴウ	ジョウホウギジュツカイハツカブシキガイシャ	カブシキガイシャマジカルムーブ	アキタステーションビルカブシキガイシャ	カブシキガイシャカネコカイソウテン	<NA>	<NA>	<NA>
NYS_CNV_P_NAME_KANA__nfkc_strip_legal_form_kana_fold_l	ソテイアツクキコウテレヒセイサクカイシヤキコウ	シヨウホウキシユツカイハツ	マシカルムウフ	アキタステウシヨンヒル	カネコカイソウテン	<NA>	<NA>	<NA>
PERSON_NAME_l	<NA>	<NA>	<NA>	<NA>	<NA>	ウツギ　チカゴ	イノウエハル	オオイシルミナ
PERSON_NAME__kana_fold_l	<NA>	<NA>	<NA>	<NA>	<NA>	ウツキ　チカコ	イノウエハル	オオイシルミナ
PHONE_NUM_4_l	9687	7503	1854	6636	7842	6766	3064	8921
POSTAL_l	3170062	5060012	3580012	0292204	5440021	1820013	8160844	5690833
SOURCE_ADDR_CD_l	<NA>	9695438217	<NA>	3393592139	2532423942	5158190801	5683835425	<NA>
ADDRESS_r	茨城県日立市平和町1-18-29	<NA>	埼玉県入間市東藤沢3-5-20	岩手県陸前高田市気仙町29-19	<NA>	東京都調布市深大寺南町3-20-6	福岡県春日市上白水4-12-28	大阪府高槻市唐崎南1丁目5番7号
ADDRESS__address_r	日立市平和町1-18-29	<NA>	入間市東藤沢3-5-20	陸前高田市気仙町29-19	<NA>	調布市深大寺南町3-20-6	春日市上白水4-12-28	高槻市唐崎南1-5-7
CIF_ID_r	8513027962782	8676088365959	6827611087602	4433203496731	2262552829629	8735172074052	6215252974995	6155805329141
FULL_NAME_KANA_r	<NA>	<NA>	<NA>	<NA>	<NA>	カネダチカコ	タニグチハルナ	ウノルミ
GIVEN_NAME_KANA__kana_fold_r	<NA>	<NA>	<NA>	<NA>	<NA>	<NA>	ハルナ	ルミ
NYS_CNV_P_NAME_KANA_r	ニッシンカサイカイジョウホケンカブシキガイシャ	チイキシンブンシャカブシキガイシャ	カブシキガイシャフコクセイメイホケン	ケイオウフェアウェルサポートカブシキガイシャ	カブシキガイシャカネコカイソウテン	<NA>	<NA>	<NA>
NYS_CNV_P_NAME_KANA__nfkc_strip_legal_form_kana_fold_r	ニツシンカサイカイシヨウホケン	チイキシンフンシヤ	フコクセイメイホケン	ケイオウフエアウエルサホウト	カネコカイソウテン	<NA>	<NA>	<NA>
PERSON_NAME_r	<NA>	<NA>	<NA>	<NA>	<NA>	カネダチカコ	タニグチハルナ	ウノルミ
PERSON_NAME__kana_fold_r	<NA>	<NA>	<NA>	<NA>	<NA>	カネタチカコ	タニクチハルナ	ウノルミ
PHONE_NUM_4_r	9687	7503	1854	6636	8829	6766	3064	8921
POSTAL_r	3170062	5060012	3580012	0292204	5440021	1820013	8160844	5690833
SOURCE_ADDR_CD_r	1923269550	<NA>	5391676869	<NA>	<NA>	5158190801	5683835425	8243032208

Why a pair matches: evidence in bits

The prior (how likely two random records are the same entity) starts the score; each comparison then adds the bits log2(m/u) of the level the pair lands in, or 0 when a side is missing. Shown for the accepted true match with a typo closest to the threshold, and for the rejected non-match closest to it.

In [11]:
```
model = fit(FINAL, train_records, train_truth, "train")
shown = res.records.set_index(uid)[list(EXPLAIN_SHOW)]
left, right = shown.loc[res.scored[l]].set_index(res.scored.index), shown.loc[res.scored[r]].set_index(res.scored.
index)
filled = (left[list(EXPLAIN_FILLED)].notna() & right[list(EXPLAIN_FILLED)].notna()).all(axis=1)
differs = left[EXPLAIN_FILLED[0]].ne(right[EXPLAIN_FILLED[0]]).fillna(False)
examples = {"accepted true match with a typo": res.scored[accepted & ~res.scored["fp"] & filled & differs].nsmalle
st(1, "match_weight"),
            "rejected non-match closest to the threshold": res.scored[~accepted & res.scored["fp"]].nlargest(1, "m
atch_weight")}
for title, pair in examples.items():
    row = pair.iloc[0]
    print(f"{title}: {row['match_weight']:.2f} bits → match probability {row['match_probability']:.4f} · found by
{row['sources']}")
    display(shown.loc[[row[l], row[r]]])
    display(waterfall(model, {c.name: row[f"gamma_{c.name}"] for c in FINAL.model.comparisons}, len(res.records)).
round(2))
```

```
accepted true match with a typo: 6.61 bits → match probability 0.9899 · found by address+address_code+tfidf_person_n
ame
```

CONT_ID	FULL_NAME_KANA	NYS_CNV_P_NAME_KANA	ADDRESS	PHONE_NUM_4	CIF_ID
0000003277	マツモトヒサオ	<NA>	福島県福島市吉倉4-11	0599	5227073594045
0000032891	マツモトヒサウ	<NA>	福島県福島市吉倉4-11	0499	5912064752971

	evidence	level	bits
0	prior	two random records match	-14.98
1	name	fold JW low	3.98
2	company	missing	0.00
3	address	same address	14.01
4	phone	one digit off	4.50
5	cif	else	-0.90

```
rejected non-match closest to the threshold: 2.64 bits → match probability 0.8614 · found by address+address_code
```

CONT_ID	FULL_NAME_KANA	NYS_CNV_P_NAME_KANA	ADDRESS	PHONE_NUM_4	CIF_ID
0000019441	<NA>	セルシスカブシキガイシャ	東京都多摩市連光寺5-18-5	8813	2754477213332
0000020498	ヤマモトカヨ	<NA>	東京都多摩市連光寺5-18-5	6813	4590715084681

	evidence	level	bits
0	prior	two random records match	-14.98
1	name	missing	0.00
2	company	missing	0.00
3	address	same address	14.01
4	phone	one digit off	4.50
5	cif	else	-0.90

Output: train / val / test → artifacts and gold (test is the final report)

Writes accepted pairs, the cumulative clusters and the unresolved records per dataset, plus `medallion/gold/<dataset>.csv` (CONT_ID → cluster). The counter flags configurations scored on test more than once.

In [12]:
```
summary = []
for dataset in DATA.datasets:
    res, records = run(FINAL, dataset), load(DATA, dataset)[0]
    acc = res.scored[res.scored["match_weight"] >= FINAL.min_match_weight].assign(stage=FINAL.stage, rule=FINAL.mo
del.name)
    pairs = acc[[l, r, "stage", "rule", "match_weight", "match_probability", "sources"]].assign(score=acc["match_p
robability"], kept=True)
    resolved = res.records[[uid, "CLUSTER_ID", "MATCH_STAGE", "MATCH_RULE"]].set_axis([uid, "cluster_id", "resolve
d_stage", "resolved_by"], axis=1)
    unresolved = res.records.loc[res.records["MATCH_STAGE"].isna(), list(records.columns)]
    assert len(unresolved) + resolved["resolved_stage"].notna().sum() == len(records)
    write_stage(DATA, dataset, FINAL.stage, pairs, resolved, unresolved)
    write_gold(DATA, dataset, resolved)
    m = res.metrics.iloc[-1]
    append_leaderboard(DATA, "probabilistic:FINAL", FINAL, dataset,
                       {**m.drop("stage").to_dict(), **{f"stage_{x}": v for x, v in res.stage_metrics.items()}})
    summary.append(dict(dataset=dataset, records=len(records), recall_before=res.metrics.iloc[0]["recall"], **m[SH
OW[1:]].to_dict(),
                        accepted=len(acc), seconds=round(res.seconds, 1)))
```

```
display(pd.DataFrame(summary).round(4))
display(test_evaluations(DATA))
```

	dataset	records	recall_before	TP	FP	recall	recall_lo	recall_hi	fp_rate_hi	pairs_FP	overmerged	dup_unresolved	accepted	seconds
0	train	30147	0.6923	6194	0	0.8928	0.8853	0.8998	0.0005	0	0	1206	1334	5.6
1	val	10002	0.6933	2037	0	0.9147	0.9024	0.9256	0.0015	0	0	316	471	0.9
2	test	10066	0.7106	2126	0	0.9020	0.8893	0.9134	0.0014	0	0	346	427	0.9

	family	configs_on_test
0	deterministic	1
1	probabilistic	1

Optuna: component combinations with a cross-validated objective

Each trial switches generators off, picks the m method, term frequency, level thresholds and resolver; its acceptance threshold comes from `THRESHOLD_RULE` on its own out-of-fold scores, as `FINAL`'s does. The objectives (**minimise FP**, **maximise recall**) are computed **out-of-fold** on train, so tuning never rewards a model scored on its own fitting data. Live dashboard: `http://127.0.0.1:8081`.

In [13]:
```
def spec_for(trial: optuna.Trial | optuna.trial.FixedTrial) -> ProbSpec:
    off = {b for b in PTUNE.toggle_blockers if not trial.suggest_categorical(f"use:{b}", [True, False])}
    m_method = trial.suggest_categorical("m_method", list(PTUNE.m_methods))
    tf = {c.name: c.tf for c in SPEC.model.comparisons} if trial.suggest_categorical("term_frequency", list(PTUNE.
tf_options)) else {}
    model = replace(FINAL.model, m_method=m_method,  # tf keys come from SPEC: FINAL may have dropped them
                    comparisons=tuple(replace(c, tf=tf.get(c.name)) for c in FINAL.model.comparisons))
    for (comparison, label), (lo, hi, step) in PTUNE.level_thresholds.items():
        model = with_level(model, comparison, label, threshold=trial.suggest_float(f"{comparison}|{label}", lo, h
i, step=step))
    return replace(FINAL, blockers=tuple(b for b in GENERATORS if b.name not in off), model=model,
                   resolver=PTUNE.resolvers[trial.suggest_categorical("resolver", list(PTUNE.resolvers))])


def objective(trial: optuna.Trial) -> tuple[float, float]:
    spec = spec_for(trial)
    try:
        weight = threshold_of(spec)
```

```
    except OverBudget as e:
        raise optuna.TrialPruned(str(e))
    at = cv_threshold_curve(cv(spec), spec, train_prev_pairs, train_truth, FOLDS, DB, DATA, [weight]).iloc[0]
    for x, v in dict(min_match_weight=weight, recall_lo=at["recall_lo"], recall_hi=at["recall_hi"],
                     candidates=summed(cv(spec))["candidates"]).items():
        trial.set_user_attr(x, plain(v))
    return float(at["FP"]), float(at["recall"])


storage = start_dashboard(PTUNE)
print(f"study: O(T · k · (fit + candidates + score + resolve)), T = {PTUNE.n_trials}, k = {CV_FOLDS}; "
      f"each trial re-samples u only for the comparisons it changed (u is cached per comparison and fold)")
study = new_study(PTUNE, storage, ["FP", "recall"], ["minimize", "maximize"])
study.optimize(objective, n_trials=PTUNE.n_trials, show_progress_bar=True)
```

```
Optuna dashboard: http://127.0.0.1:8081  (storage sqlite:////Users/marvinma/projects/entity matching/data/artifacts/
optuna/probabilistic_matching.db)
study: O(T · k · (fit + candidates + score + resolve)), T = 30, k = 3; each trial re-samples u only for the comparis
ons it changed (u is cached per comparison and fold)
study 'probabilistic_20261104_161741' → watch it live in the dashboard
```

```
Bottle v0.13.4 server starting up (using WSGIRefServer())...
Listening on http://127.0.0.1:8081/
Hit Ctrl-C to quit.
```

Pareto front, re-scored on val

The suggested point is the cheapest configuration within the noise of the best 0-FP val recall (`pick_within_noise` ; none if no point has 0 FP on val). It is judged by the same rule as `DECISIONS` ("better"), against `FINAL` ; to adopt a passing point, add it as a `CHANGES` + `DECISIONS` entry.

In [14]:
```
front, seen = [], set()
for t in sorted(study.best_trials, key=lambda t: (t.values[0], -t.values[1])):
    spec = replace(spec_for(optuna.trial.FixedTrial(t.params)), min_match_weight=t.user_attrs["min_match_weight"])
    if repr(spec) in seen:
        continue
    seen.add(repr(spec))
    v = run(spec, "val").metrics.iloc[-1]
    front.append(dict(trial=t.number, cv_FP=t.values[0], cv_recall=t.values[1], val_FP=v["FP"], val_recall=v["reca
```

```
ll"],
                      val_recall_lo=v["recall_lo"], val_recall_hi=v["recall_hi"], val_zero_fp=v["FP"] == 0,
                      candidates=t.user_attrs["candidates"], min_match_weight=t.user_attrs["min_match_weight"],
                      term_frequency=t.params["term_frequency"], m_method=t.params["m_method"], spec=spec))
front = pd.DataFrame(front)
display(front.drop(columns="spec").drop_duplicates(["val_FP", "val_recall", "candidates"]).round(4))  # one row pe
r distinct outcome
best = pick_within_noise(front, "val_recall", "val_recall_lo", "val_recall_hi", "val_zero_fp", "candidates") if le
n(front) else None
if best is None:
    print("no Pareto point has 0 FP on val: FINAL stays")
else:
    spec = front.at[best, "spec"]
    on_cv = paired_test(joined(cv(FINAL)), joined(cv(spec)), train_true)
    on_val = paired_test(joined(run(FINAL, "val")), joined(run(spec, "val")), val_true)
    print(f"suggested: trial {front.at[best, 'trial']} → adopt by rule ('better' vs FINAL): {adopt('better', on_c
v, on_val, DECISION_P)}")
    print(f"cross-validated train {on_cv} · val {on_val}")
```

	trial	cv_FP	cv_recall	val_FP	val_recall	val_recall_lo	val_recall_hi	val_zero_fp	candidates	min_match_weight	term_frequency	m_method
0	19	0.0	0.9076	0	0.9151	0.9028	0.926	True	29675	6.0	False	labels

```
suggested: trial 19 → adopt by rule ('better' vs FINAL): False
cross-validated train {'TP_gains': 11, 'TP_losses': 7, 'p': np.float64(0.480682373046875), 'FP_added': 0, 'FP_remove
d': 0} · val {'TP_gains': 5, 'TP_losses': 4, 'p': np.float64(1.0), 'FP_added': 0, 'FP_removed': 0}
```